S(22): A COMPLETE COMPUTER-ASSISTED DETERMINATION

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Theorem. For elliptic curves over number fields of exact degree22,
S(22)={2,3,5,7,11,13,17,19,23,29,31,37,41,43,47,53,67}.
All required local, character, independent replay and coverage certificates have passed. There is no unresolved case.

1. STATEMENT, OCCURRENCES, AND TRACE DIVISORS

The lower bound is explicit. Small rational examples for2,3,5,7 are extended to the Eisenstein degree22 field defined by t^22-2. The seven verified degree11 examples for11,13,17,19,23,29,31 are independently extended by sqrt(2); the resulting fields have absolute degree22 and the points still have the asserted exact prime orders. Direct degree22 examples for37,41,43,47,53,67 are recorded in the packet. Their field polynomials are irreducible and the displayed point P is nonzero with pP=0 in exact arithmetic. In particular, the53 example uses a quadratic extension of the degree11 x-coordinate field. For67, an explicit CM construction on a twist of y^2=x^3+1 gives an irreducible degree22 polynomial and a point verified by direct multiplication. The native occurrence checks are fresh; no claim about a degree11 proof file being pinned is needed.

Conversely, if p divides the torsion order, Cauchy’s theorem gives a K-rational point of exact orderp. Its point on X=X1(p) has degree r with r|22. We use D=(22/r) times its rational trace, an effective rational noncuspidal divisor of degree22. Its geometric support is Galois transitive. Ramified reduction and repeated points are allowed.

We do not use the odd-degree real-diamond argument. A diamond of orderh on a characteristic-zero noncuspidal point has orbit lengthh, h/2, orh/3 when the indicated divisor exists. The shorter orbits occur only at the appropriate CM points. If a rational diamond preserves D, its orbit length divides the degree r of the underlying closed point and hence divides22. The action on cusps is free.

2. THE IMPROVED DYADIC BOUND

Let J=J0(p), e its winding element, I=Ann_T(e), and A=J/IJ the optimal winding quotient. Its rational points are finite. Write
M=H0(X0(p)_Z2,Omega)[I], L=pi^*Cot(A_Z2).
The integral coefficient pairing identifies M with the Z2-dual of T/I. Raynaud’s cotangent bound gives 2M subset L subset M. Equality is not assumed. The precise earlier lemma and its references are retained in references/dyadic_theorem_inherited.tex. It follows by applying Raynaud’s bound to the optimal quotient at ramification index1=2-1; the saturation of L is M and its quotient by L is killed by2.

Set m=44 and M0=229. Let epsilon be the indicator of residues115,...,228. For every u=1,...,228 the matrix
(epsilon(na)-epsilon(nu/a))_(1<=n<=44,1<=a<=228)
has rank44 overF2; all arguments are modulo229. Every entry and rank is rebuilt independently in Sage. The DKSS integral intersection formula, with p*u=1 modulo229 and p>2*44*229=20152, identifies these rows with the intersections of (T’_(2n)-2T’_n)e with {0,a/229}. Here
T’_n=sum_(a|n)mu(n/a)T_a.
These elements span the same integral module as (T2-3)T_n e, n<=44. Moreover (T2-3)T e lies in integral absolute homology. Thus a dependence

between T_1 e,...,T_44 e modulo2T e would contradict the intersection matrices. Duality gives coefficient surjectivity on M and cotangent image containing2Z2^44 for the degree44 Abel map to A at44 times a cusp.

The torsion in the rational dyadic reduction kernel is killed by2. If a degree22 divisor specializes to22 times that cusp, its elementary symmetric polynomial is1+2B(t). The doubled divisor has polynomial 1+4(B+B^2), so it lies in the formal neighborhood modulo4. The winding image of the doubled divisor is0. Pullback functions with linear terms 2x_i and higher integral terms of degree at least2 cannot vanish at a nonzero vector in4Z2^44: if k>=2 is the minimum coordinate valuation, one linear term has valuationk+1<2k. Hence the formal zero fiber is only the doubled cusp. This excludes such a noncuspidal trace.

An ordinary elliptic curve overF_(2^f) has even group order. At f=22 the Hasse upper bound is4198401, so its largest possible even group order is 4198400. An odd prime divisor is at most2099200. All smaller ordinary residue degrees give smaller bounds. The degree22 supersingular orders are

4190209,4192257,4194305,4196353,4198401.

Their largest prime factor is599479, from4196353=7*599479. All seven supersingular twists are counted natively and independently. The earlier complete lists through degree21 are retained and checked through the frozen parent packet. At degree17 the factors are137*953,3*43691, 5*26317, all below100000. At f<=16 even the Hasse upper bound is below 100000. Both toric orders2^f-1 and2^f+1 for every f<=22 are included. Additive reduction cannot support these large odd primes.

For p>2099200, every dyadic place must therefore be multiplicative with P on a nonidentity component. The image on X0(p) is the same rational cusp at every place; after Atkin--Lehner it is infinity. The preceding dyadic argument excludes this. Thus p<=2099200. A conservative screen using every ordinary even multiple, every supersingular order, and both toric orders for all f<=22 leaves1488 primes above100000. No residue degree is assumed to divide22; arbitrary partitions at the dyadic places are allowed.

3. GLOBAL DIAMOND AND CHARACTER EXCLUSIONS

We use Derickx--Stoll, arXiv:2505.14109v1, Theorem3.3, Proposition4.1, Remark4.4, Corollary7.5, and Table2 only within its proved range p<100000. The gonality bound is

gon_Q(X1(p))>=325*(p^2-1)/65536.

If p^2>(2^19/325)*22+1 and (p-1)/(2str(p)) does not divide22, Corollary7.5 excludes p, subject to its CM exceptions. Apply it to the actual point degree r|22. The only prime CM exceptions for even r|22 are5,7,67, all verified occurrences. The published reductions leave

59,61,71,73,79,83,89,97,101,103,107,109,113,127,131,137,139,
163,167,173,181.

The first seventeen and163 are treated locally.

At167, diamond2 has order83. The inherited exact odd A3 determinant on its full anti homology has dimension2132. This gives the required 2-primary control, and Proposition4.1 applies with5*22=110 below the gonality bound. Every geometric orbit has length83, which does not divide22.

At173, diamond4 has order43. Fresh exact A3 determinant norms on character orders43,86, of full rational homology dimension2268, are odd.

The same norms are independently reconstructed with Sage modular symbols. Hence A3 is an odd-degree isogeny on the anti part. Its finite rational image has no2-primary torsion. Proposition4.1 applies with 6*22=132 below the gonality bound, forcing order43 invariance, impossible in degree22.

At181, put r=diamond2, h=r^6, of order15. Its only positive nontrivial character order is6, so (h-1)J(Q) is finite. The inherited independently checked anti homology has dimension2388. Put u=r^2, of order45, and
E=5*N_(u^5)-N_u, beta=E*A11+(45-E)*A3,
where N denotes the sum of powers. On orders5,10, E is45; on the other anti components E is0. The corresponding A11 orA3 determinant norms are odd. Thus beta kills rational torsion and is an odd-degree isogeny on the anti part. Proposition4.1 with6*22=132 applies. Orbit lengths15,5 both fail to divide22.

At163 the old6d argument is not applicable:6*22=132 is not strictly below325*(163^2-1)/65536. This level is treated locally on a fresh model, not excluded by that failed inequality.

For the1488 large primes, choose a prime r with v_r((p-1)/2)>v_r(22). The selected orders are exactly those divisors n of (p-1)/2 with
v_r(n)>=v_r((p-1)/2)-v_r(22).
Every character of every selected order is shown nonstrange. This bounds the r-part of str(p), so the quotient (p-1)/(2str(p)) has r-valuation strictly larger than v_r(22). Corollary7.5 then excludes p.

There are10065 selected orders:5018 fresh and5047 certified inherited orders. For each fresh order, an exact finite-field Manin presentation is exported. Its two-term quotient, all three-term relations, weighted dual graph, and four-term T2 operator are retained. Independent forest and root certificates prove its exact rank. If k is its cuspidal sign component dimension, the full relative space has dimensionk+2. The starting vector is the winding element multiplied by both Eisenstein factors T2-1-2chi(2) and T2-chi(2)-2; it lies in the cuspidal space.

The exact sparse integer engine generates2k scalar Krylov terms. A rational reconstruction with denominator degreek, numerator degree at mostk-1, and coprime numerator and denominator proves scalar linear complexityk, hence a k-dimensional winding orbit. The full congruence, coprimality and degrees are independently checked with FLINT, separately from the NTL producer. All primitive characters of a fixed order are Galois conjugate. Nonvanishing after one exact reduction proves the required characteristic-zero nonstrangeness.

The packed engine uses unsigned integer arithmetic only. Its reciprocal reduction is exact: for mu=floor(2^64/q), floor(z*mu/2^64) differs from floor(z/q) by at most1 for0<=z<2^64. One correction suffices. Every accumulation is bounded before selecting the64-bit path, with128-bit fallbacks. All reference terms and adversarial graph fixtures, including a modulus4294967291, agree with the earlier exact implementation. Every job binds its graph, forest, native prefix, full sequence, recurrence, source and binary hashes. No floating-point rank or heuristic conclusion is used.

Some jobs evaluate eight independent sparse rows in parallel. This is also unsigned integer arithmetic only. If z<2^B, with B>=32 and mu=floor(2^B/q)<2^32, writing z=2^32*zh+zl gives the exact estimate
floor((zh*mu+floor(zl*mu/2^32))/2^(B-32)).

It underestimates floor(z/q) by at most1. The intermediate integer fits 64 bits, and the final multiply fits its asserted32-bit factors. One integer comparison and subtraction corrects the remainder. The scalar and128-bit paths remain available when these bounds fail. All349866 terms of the largest reference and all175000 terms of the earlier reference agree byte for byte, as do21 independent dense graph fixtures. A further7200000 integer reductions agree with direct remainders. The launcher handoff drains existing children and preserves their actual engine records; completed work is retained without mixing provenance.

The independent prime ledger contains all155750 primes through2099200.

4. FINITE-FIELD INPUTS AND PRINCIPAL-DIVISOR CERTIFICATES

The source census includes all ordinary j-fields and twists, including proper j-fields and quadratic nondescent twists, and all p-torsion lines. At89 full89-torsion occurs overF_(2^22); all90 cyclic subgroups are included. Independently computed binary-quadratic-form class numbers include the correction p*H((t^2-4q)/p^2) for full torsion. Supersingular orbits are independently reconstructed with the quaternion automorphism group. Frobenius traces at89 are additionally recomputed with Sage modular symbols through degree22. The census at163 is fresh.

Integral saturated canonical q-bases and modular-unit relations locate the actual elliptic points on the canonical curve. All unit relations are certified through orders exceeding their global section-degree bounds. In the fresh genus1027 model at163, precision4628 and pole bounds348,364 are independently rebuilt from all rational cusp valuations and all nonrational cusp values. The unit relations and the point-diamond action are independently checked using Tate series.

Every native Hecke target is an actual elliptic quotient by a cyclic kernel. All division-polynomial factors are included with their correct field and trace multiplicities. Points remain nonzero p-torsion. Hensel lifts and canonical-coordinate jets satisfy all curve, division and unit relations through the retained precision. Signed divisor multiplicities are combined as integers before taking positive and negative parts. Even multiplicities are never discarded modulo2.

Here is the principal-divisor certificate. Let Z=Zplus-Zminus be the required degree-zero divisor on the special curve and let V=H0(L), where L is the canonical or bicanonical line bundle. Choose nonzero F and several H in H0(L-Zplus). If Z were principal, say div(f)=Z, then v=F/f would be a nonzero element of H0(L-Zminus), and H*v would belong to F*V for every chosen H. These are necessary homogeneous linear conditions on v. Their combined full rank dim(V) proves that Z is not principal. All multiplication tensors are rebuilt from q-series through an order exceeding the Sturm or section-degree bound. Every retained obstruction matrix is independently ranked in Sage. Random choices only find certificates; each certificate is exact and deterministic to check.

The auxiliary section space is bicanonical at61 and97. At97 the canonical dimension345 is smaller than a possible positive degree352. The original basis is extended from precision1768 to3500, preserving all earlier coefficients, diamond and unit matrices. Independent basis calibration checks every entry. Bicanonical dimension1032 and product dimension2408 are verified, with weight8 Sturm bound3136<3500. The initially insufficient canonical attempt is unused.

Relative families and stabilizer arguments below are applied to the exceptional source orbits left by the inherited direct obstructions. Other source orbits of the same degree remain covered by those direct certificates. New mixed supports are checked separately whenever their added blocks are not killed by the inherited annihilator.

New degree22 ordinary singles, all ordinary pairs with total22, and all triples7+7+8 are covered. Triples7+7+7 and lower pair totals are inherited with their degree-independent principal-divisor certificates. At163 all ordinary sources through22 and all163 relative positions of the two degree10 orbits are checked freshly. Independent Burnside counts verify the new unordered triple enumeration. The input and coverage checks identify every native source and every relative pair position.

5. INTEGRAL RELATIVE FORMAL IMMERSIONS

We use actual integral endomorphisms with finite rational image and outer torsion annihilators. At odd q, A_q annihilates rational torsion by good reduction. For q=2, the proved rational cuspidal2-primary result is used only at levels<=157. Above that range we use explicit odd anti isogenies, not a conjecture about the whole rational torsion group.

The cotangent calculations use actual saturated canonical lattices or odd-order quotient invariants. At a free ordinary H8 orbit, the local quotient is etale even at2; no saturation assertion for a global even quotient is used. Pure cusp tests on modular symbols use only the implication that operator independence on an integral stable lattice implies independence in the integral Hecke order. Injective compression is verified by reconstructing every raw row.

The relative formal lemma is integral. If a morphism vanishes on a smooth formal family L of dimensionr inside a smooth formal space of dimensiond and has normal cotangent rankd-r, its pullback functions generate the ideal of L. Thus its zero locus is exactly L. There are no integration denominators. Closed-point coefficient and trace-pairing matrices differ by invertible changes of basis, also in even residue degree. Symmetric-power jets use coefficients without factorials.

The standard relative cases, written(base degree,base rank,cusp degree), are
 61 ordinary20:(20,16,2), all6 translates;
 61 SS15:(15,12,7);61 SS20:(20,16,2);
 73 SS6:(6,6,16);73 NC9:(9,8,13);
 73 doubleSS6:(12,12,10), all10 unordered positions;
 73 SS6+NC9:(15,14,7), all16 positions;
 97 SS16:(16,16,6), with sharper finite Hecke maps;
 109 SS9:(9,8,13);113 ordinary8:(8,7,14).
Every matrix entry is independently rebuilt. Complete relation spaces on dominating multiplicity profiles cover all cusp multiplicities. Independent integer partitions and placements check completeness. The cost of a relation is the sum of prefix lengths at the cusps.

A zero family consists of free pullbacks, tame fixed-point lifts, and a fixed residual cusp divisor. A nonzero fixed cusp remainder contradicts noncuspidal support. At97, where cusp6 can be two full H3 fibers with no remainder, the SS16 points lift to H3-fixed points while points in cusp disks are not H3-fixed. This Galois-invariant property contradicts transitivity of the geometric support. This argument applies in even degree.

Pure cusp immersions pass at every local level. At103 and131 the odd quotients leave only their known full cusp fibers of degrees17 and13. The intrinsic relative ranks are21 in all20349 and293930 configurations, respectively; the fixed remainders have degrees5 and9. At163 use h=diamond8, of order27. The inherited full anti A3 determinant norms are odd, so the actual cotangent image of(h-1)A3 equals(h-1)Omega modulo2. All241 dominating cusp profiles are independently checked. The only relations come from fibers of degree27, too large for degree22.

6. THE NEW LEVEL89 QUADRATIC QUOTIENT

Put H=H11 and Y=X1(89)/H. The H-action is free, including at cusps. The quotient is smooth at2,3 and has genus27; odd-order averaging gives the full invariant cotangent lattice. The inherited two-sign anti A3 norms show that im(h-1)J(Q) has no2-primary torsion.

For an H-invariant special noncuspidal block, choose a rational cusp divisor C lifting its rational-cusp complement. The class(h-1)[D-C] is rational torsion and reduces to0, hence is0. Its principal divisor has positive degree at most22+deg(C)<=33, below the upstairs gonality bound. Separating cuspidal and noncuspidal supports forces hD=D and hC=C. Thus D is the pullback of a rational noncuspidal degree2 divisor D’ on Y.

Independent Frobenius traces give #Y(F2)=8 and #Y(F4)=38. The8 rational points are cusps, and the remaining30 geometric points form15 SS points of degree2. A map Y->P1 overF2 must have degree at least8, since 38<=5d. Good-reduction specialization gives gon_Q(Y)>=8.

For the NC11 configurations, D’ specializes to two cusp points. Actual maps(D-1)(Tq-Dq-q)(Tq-1-qDq) have finite rational image and kill all cusp differences. Every one of the36 unordered cusp pairs has cotangent rank2. The invariant forms, Hecke operators and Tate jets are independently rebuilt, so formal immersion excludes these noncuspidal D’.

The five SS22 source orbits have lengths4,4,4,2,1. Source4763 is excluded by T3. Sources4764,4765 are excluded by T5, with T7 as an additional check. The relevant deformation coefficients and isogenies are independently replayed and the obstruction matrices have rank287.

For the last two orbits, the residual C4 diamond involution iota stabilizes the special D’. Choose an invariant rational cusp divisor C2=c+iota(c). The class[iota(D’)-D’]=(iota-1)[D’-C2] is rational torsion, since Table2 at89 has no positive nontrivial character component. It reduces to0, so its double is0. A nonzero principal divisor 2*iota(D’)-2*D’ would have positive degree at most4<gon_Q(Y). Therefore iota(D’)=D’.

All four points fixed by iota form a single degree4 Galois orbit. Here is an exact check at the good prime3. On E:y^2=x^3-x, Aut(E) has order12, generated by i:(x,y)->(-x,sqrt(-1)y) and u:(x,y)->(x+1,y). Frobenius commutes with u and anticommutes with i; trace0 and norm3 identify its coset with plus or minus j=2u+1, with j^2=-3. The unique split quaternion representation on E[89] gives60 nonzero-vector orbits under Aut(E)*H11. Exact F89 matrices show that iota fixes four of them and Frobenius is one4-cycle on those four. The tame fixed locus is finite etale at3, so this is also its characteristic-zero Galois action. Every fixed point is accounted for, since a characteristic-zero fixed point has j=1728.

A degree2 rational divisor cannot contain a degree4 point. Thus an iota-invariant D’ is a free pair and descends to a rational noncuspidal point on Y/<iota>, which maps to X0(89). Mazur’s prime-isogeny theorem excludes such a point. This resolves the remaining SS22 disks. Detailed matrices, permutations and the argument are retained in argument89.txt and its exact certificates.

7. NEW MIXED SUPPORTS AT113 AND127

At113, the35 ordinary closed points of degree8 form two diamond orbits, of lengths7 and28. The length7 orbit is H8-invariant; the other orbit needs a separate argument. These counts and both actual diamond orbits are independently checked. The generic orbit is represented by source3, and the H8 orbit by source2.

For the generic orbit, apply(D-1)A3 to the whole mixed divisor. Its stabilizer H2 acts trivially on the four SS7 closed points and the four NC14 closed points. Fixing the ordinary reference therefore leaves four NC14 positions and ten unordered SS7 pairs, allowing repetitions. All14 combined principal-divisor obstructions have independent rank477.
On a nonrational cusp orbit C, T3(C)=3C+delta3(C), so
A3(C)=2(C-delta3(C)).
The cusp jets are constructed on the nodal model y^2+xy=x^3 at the 113th-root point and deformed by a6=u. Its discriminant is u and j=1/u, so u is a cusp uniformizer. Eight canonical jets are checked independently with Sage; all descend from F_(2^28) to F_(2^14). The literal diamond and delta3 orientations are checked against the113th root. Thus these cusp multiplicities are retained as integers, including the factors2.

For the exceptional H8 orbit, O8+SS7+SS7 has total22. Apply(D^7-1)A3. The ordinary block is H8-invariant as a closed divisor, so its entire Hecke contribution is0. The three unordered SS-pair relative positions 0,1,2 give nonprincipal divisors, with independently verified rank477 matrices. The independent quaternion enumeration gives exactly four SS7 closed points in one diamond orbit, proving completeness. All used SS jets and T3 isogenies are independently rebuilt. No wild involution lifting argument is used.

For the exceptional H8 orbit in O8+NC14 use the actual double-Eisenstein maps(D^7-1)AqAq’. The
NC point is computed with a113th root inF_(2^28), since2^14=-1 modulo113; its canonical coordinates descend toF_(2^14). The explicit subfield embedding is retained and independently checked. The ordinary block has rank7, the NC block rank14, and their combined rank is21. Thus the zero formal family is an H8 ordinary pullback plus a fixed degree14 cuspidal divisor, contradicting noncuspidal support. Closed orbit lengths 7 and4 are coprime, so the representative covers every relative position.

The support O8+SS7+C7 is excluded by forced H2-invariance of the odd cusp complement:2*(22+7)=58 is below the gonality bound, and an invariant cuspidal divisor has even degree. For2*O8+C6 the inherited pair certificates leave only pairs from the
exceptional H8 orbit. Their forced H8 orbit lengths8,4 do not divide22. The pure degree14 and21 SS/NC configurations have
H14 orbit lengths14,7, neither dividing22; their new doubled bounds are strict.

At127, O9+NC7+C6 has two free pullback directions. Six cusp jets give

a378-column binary relation code. Its full relation dimension is90 and each jet level is the7-by9 row/column code of minimum nonzero weight7. Thus every choice of six prefix columns is independent modulo the base, covering109453344 configurations. The total normal rank is20 and the fixed cusp complement has degree6. Both the full kernel and every intrinsic matrix entry are independently checked.

The new supports O8+SS14 and O8+2*NC7 at127 reuse the ordinary O8 obstruction: the added blocks are fixed by diamond2 as divisors. The new O7+SS15 support at61 similarly reuses its unique degree7 ordinary (D^6-1)A3 obstruction, because SS15 is H5-invariant.

8. THE REMAINING STABILIZERS AND COMPLETE COVERAGE

At73, nonordinary totals18 and21 are H9-invariant. The inherited actual rational2-torsion computation, with diamond2 trivial on its upper bound, gives the undoubled class relation. Since H9 has odd order, a2-torsion class in its anti image is0. The bounds22+C<=26 are strictly below the gonality bound. Orbit lengths9,3 do not divide22.

At109 use H18 on nonordinary totals18,21. Their doubled bounds52,46 are strict, and orbit lengths18,9,6 do not divide22. At113 use H14 as above. At127 all pure NC7/SS14 configurations are excluded by H7-invariance; the largest doubled bound is2*(22+15)=74, below the gonality bound. H7 acts freely and7 does not divide22. At137 the required H17 orH34 invariance has doubled bound54; orbit lengths17,34 do not divide22. The ordinary-pair exceptional H9 at127 has orbit lengths9,3. These are all characteristic-zero orbit arguments and do not invoke odd degree.

The local inventory contains514 support types. A second enumeration uses Mobius inversion of independent class-number point counts, quaternion supersingular orbits, the multiplicative order of2, and all Sage integer partitions with every type assignment. It gives exactly the same514 cases. Every case is bound to an exclusion certificate. All new binary obstruction matrices, all complete cusp relation spaces, all raw/compressed operator rows and all relevant native input sets are checked independently. Completed native chunk archives are lossless: each decompressed stream is rehashed against the original bytes before the raw chunk is removed. Combined proof inputs remain directly readable.

All required checks have passed. The finite prime ledger and dyadic bound exclude every prime outside the displayed lower set. This completes the computer-assisted determination. There is no unresolved case, no BSD or GRH hypothesis, and no conjectural rational-torsion premise. This packet does not claim a separate-agent audit of the complete degree22 proof.

PRIMARY REFERENCES AND REPLAY

Derickx--Stoll, Prime order torsion on elliptic curves over number fields, PartI: Asymptotics, arXiv:2505.14109v1 (Theorem3.3, Proposition4.1, Remark4.4, Corollary7.5, Table2 within p<100000).
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Abbes--Ullmo, A propos de la conjecture de Manin pour les courbes elliptiques modulaires, Compositio Math.103 (1996),269--286, AppendixA, TheoremA.1 (Raynaud); Agashe--Ribet--Stein, The Manin constant, Pure Appl. Math. Q.2 (2006),617--636, Section4.4.
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Theorem1 (no rational89-isogeny). The primary paper and its hash are retained.
Van Hoeij, Low degree places on the modular curve X1(N), arXiv:1202.4355,
and its explicit place data (used only to obtain witnesses checked anew).

The complete source, data, replay instructions, dependency manifest,
prime ledger, support ledger and independent-check index accompany the
proof. The inherited closure contains100184 frozen files; its manifest
SHA256 is1ce61f220611ed348af3b0e92554a21f46d2bedd48058d45b654ac3f26e8f1c5.